SAN FRANCISCO STATE UNIVERSITY

SCHOOL OF ENGINEERING

Final Project Report: LM21215 Buck Converter

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By

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1. Overview

In this final project, we were tasked with designing, simulating, building, and testing a 5 V to 1.8 V DC-DC buck converter using the Texas Instruments LM21215 synchronous buck regulator. The converter was required to supply up to 5 A of load current while meeting various electrical specifications, including:

1. Less than 30% inductor ripple current,
2. Less than 1% peak-to-peak output voltage ripple at full load,
3. Less than 50 mV peak-to-peak input voltage ripple,
4. Greater than 50° phase margin,
5. Less than 50 mV transient droop during a 2 A to 3 A load step.

The design process included creating the schematic, selecting real components, performing hand calculations, laying out the PCB, simulating the control loop and transient response, assembling the board, and measuring the final performance on the oscilloscope. The main calculated component values included the feedback resistor divider, output inductor, input capacitors, output capacitors, and compensation network. These values were chosen in accordance to the design recommendations within the LM21215 Datasheet.

In addition to electrical design, PCB layout was an important part of the project. Since switching regulators contain high-current, high-slew-rate nodes, poor layout can cause excessive ringing, noise, instability, and poor regulation. The layout therefore focused on minimizing the switched-current loop area, placing the input capacitors close to the PVIN and PGND pins, keeping the switch node copper area small, routing the feedback signal away from noisy traces, and maintaining a solid ground return path.

The final design was evaluated using both simulation and hardware measurements. SPICE simulations were used to estimate loop crossover frequency, phase margin, and transient response. Hardware testing included measuring the switch node waveform, output voltage ripple, inductor current ripple, DC output voltage, switching frequency, and transient response during 2 A to 3 A and 3 A to 2 A load steps. These measurements were used to verify whether the completed converter met the project specifications and to compare the expected theoretical performance against the actual PCB behavior.

2. Hand Calculations

To begin the design, general rough hand calculations were completed to estimate the major component values before simulation and PCB layout. These were necessary to get a rough idea

of how chosen values would eventually perform on a board, and could (and in my case, eventually were) subject to change.

The design target was a 5 V input to 1.8 V output buck converter at 5 A. The LM21215 uses a 0.6 V feedback reference and a 500 kHz switching frequency, so those values were used throughout the initial calculations. All calculations were done by setting initial values in excel, and their corresponding equations, as seen below:

Parameter	Value	Unit		Calculations	Value
Vin	5	Volts		duty_cycle	0.36
Vout	1.8	Volts		rfb1	20000
Iout_max	5	Amps		Delta_IL_target	1.25
fsw	500000	Hz		L	0.0000018432
Vref	0.6	Volts		I_pk	5.625
ripple_percent	0.25			Cout_min_ripple	1.7361111E-05
Vout_ripple_max	0.018	Volts		Cout_min_droop	0.00001152
Vin_ripple_max	0.05	Volts		Cout_min	1.7361111E-05
Istep	1	Amps		dc_with_efficiency	0.4
Vdroop_max	0.05	Volts		Cin_min	0.000048
efficiency	0.9			Icin_rms	2.4
Rfb2	10000	Ohms		Ro	0.36
Vramp	0.8	Volts		fLC	17672.88952
fc	40000	Hz		fESR	1205719.2658
Cout_eff	4.40E-05			Rc1	7242.7318608
ESR	0.003	Ohms		Rc2	297.51177854
Rds_highside	0.007	Ohms		Cc1	2.486796E-09
Rds_lowside	0.0043	Ohms		Cc2	9.1118392E-11
I_quiescent	1.50E-03	Amps		Cc3	4.4367991E-10
L_DCR	0.015	Ohms			
Cin_ESR	0.002	Ohms			
Cout_ESR_loss	0.003	Ohms			
P_sw	0.15	watts			

Losses	Values
IL_rms	5.013004
P_inductor	0.376953
P_highside_cond	0.063328
P_lowside_cond	0.069158
Ic_in_rms_check	2.4
P_cin_esr	0.01152
Icout_rms	0.360844
P_cout_esr	0.000391
P_quiescient	0.0075
Pout	9
Ploss_total	0.67885
efficiency_est	0.929863

Figure 1: Excel Calculations

			GRJ31CR71H105KE11
Reference Equations			732-7843100200CT-ND
Duty = Vout/Vin			541-294DCT-ND
Feedback r 1 = Rfb2*(Vout/Vref-1)			311-10.0KHRCT-ND
Ripple current = ripple_pct*Iout_max			541-1.00DCT-ND
L = ((Vin-Vout)*D)/(Delta_IL*fsw)			P41.2KAACT-ND
Peak Current = Iout_max + Delta_IL/2			13-RC1210FR-0720KLC
Min voltage ripple = Delta_IL/(8*fsw*Vout_ripple_max)			277-1947-ND
Minimum droop = L*Istep^2/(Vdroop_max*(Vin-Vout))			
dc output = Vout/(Vin*efficiency)			
Input Cap min current = (Iout_max*dc*(1-dc))/(fsw*Vin_ripple_max)			
Input cap RMS current = Iout_max*SQRT(D*(1-D))			
Ro = Vout/Iout_max			
LC freq = 1/(2*PI()*SQRT(L*Cout_eff))			
output cap ESR zero freq = 1/(2*PI()*ESR*Cout_eff)			
Rc1 = (fc/fLC)*(Vramp/Vin)*Rfb1			
Cc1 = 1/(PI()*fLC*Rc1)			
Cc2 = Cc1/(PI()*fsw*Rc1*Cc1-1)			
Rc2 = Rfb1*(fLC/(fESR-fLC))			
Cc3 = 1/(2*PI()*fESR*Rc2)			
RMS=SQRT(Iout_max^2 + Delta_IL_target^2/12)			
Inductor Conduction=IL_rms^2*L_DCR			
Highside conduction=duty_cycle*(Iout_max^2 + Delta_IL_target^2/12)*Rds_highside			
Lowside conduction=(1-duty_cycle)*(Iout_max^2 + Delta_IL_target^2/12)*Rds_lowside			
Input cap RMS=Iout_max*SQRT(duty_cycle*(1-duty_cycle))			
Input Capaictor RMS=Ic_in_rms_check^2*Cin_ESR			
Output Cap RMS=Delta_IL_target/(2*SQRT(3))			
Output ESR=Icout_rms^2*Cout_ESR_loss			
Quiescent Loss=Vin*I_quiescent			
Output Power =Vout*Iout_max			
Total Power=P_inductor+P_highside_cond+P_lowside_cond+P_cin_esr+P_cout_esr+P_quiescient+Psw_assumed			
Experimental Efficiency=Pout/(Pout+Ploss_total)			

Figure 2: Reference Equations

As a general overview:

*note the sections below were written in an Obsidian .md file and screenshotted because i wanted nice LATeX formatting and not whatever Google docs calls its “equation builder”

2.1 - Feedback Resistors

The output voltage is set by the feedback divider:

$$V_{out} = V_{ref}\left(1 + \frac{R_{FB1}}{R_{FB2}}\right)$$

Using:

$$V_{out} = 1.8V, \quad V_{ref} = 0.6V, \quad R_{FB2} = 10k\Omega$$

$$R_{FB1} = R_{FB2}\left(\frac{V_{out}}{V_{ref}} - 1\right)$$

$$R_{FB1} = 10k\Omega\left(\frac{1.8}{0.6} - 1\right) = 20k\Omega$$

Therefore, the selected feedback values were:

$$R_{FB1} = 20k\Omega, \quad R_{FB2} = 10k\Omega$$

2.2 - Duty Cycle

For an ideal buck converter:

$$D = \frac{V_{out}}{V_{in}}$$

$$D = \frac{1.8V}{5V} = 0.36$$

So the converter operates at approximately a 36% duty cycle.

2.3 - Inductor Selection

The target ripple current was chosen as 25% of the 5 A full-load current (less than the 30% in the spec)

$$\Delta I_L = 0.25(5A) = 1.25A$$

For a buck converter, inductance is defined as following:

$$L = \frac{(V_{in} - V_{out})D}{\Delta I_L f_{sw}}$$

$$L = \frac{(5 - 1.8)(0.36)}{(1.25)(500kHz)}$$

$$L = 1.84\mu H$$

Now that we have the inductor value, the peak inductor current is as follows:

$$I_{pk} = I_{out} + \frac{\Delta I_L}{2}$$

$$I_{pk} = 5A + \frac{1.25A}{2} = 5.625A$$

2.4 - Output Capacitor Selection

The maximum allowed output ripple was 1% of 1.8 V:

$$\Delta V_{out,max} = 0.01(1.8V) = 18mV$$

The equation for output capacitance in a buck converter is:

$$C_{out,min} = \frac{\Delta I_L}{8 f_{sw} \Delta V_{out}}$$

$$C_{out,min} = \frac{1.25A}{8(500kHz)(18mV)}$$

$$C_{out,min} = 17.36\mu F$$

Since we have a requirement of a minimum droop as well:

$$C_{out,min} = \frac{L(\Delta I_{step})^2}{V_{droop}(V_{in} - V_{out})}$$

Using:

$$L = 1.84\mu H, \quad \Delta I_{step} = 1A, \quad V_{droop} = 50mV$$

$$C_{out,min} = \frac{1.84\mu H(1A)^2}{(50mV)(5V - 1.8V)}$$

$$C_{out,min} = 11.52\mu F$$

This means that minimum capacitance needed to be above 17uF. I chose

$$C_{out,eff} = 44\mu F$$

2.5 - Input Capacitor Selection

Input Ripple Target was:

$$\Delta V_{in,max} = 50mV$$

Using the estimated efficiency of 90%, the effective input duty factor was:

$$D_{eff} = \frac{V_{out}}{V_{in}\eta}$$

$$D_{eff} = \frac{1.8}{5(0.9)} = 0.40$$

The minimum input capacitance estimate was:

$$C_{in,min} = \frac{I_{out}D_{eff}(1 - D_{eff})}{f_{sw}\Delta V_{in}}$$

$$C_{in,min} = \frac{5A(0.40)(0.60)}{(500kHz)(50mV)}$$

$$C_{in,min} = 48\mu F$$

2.6 - Compensation

The compensation network was designed for a crossover frequency of approximately:

$$f_c = 40kHz$$

smack dab in the middle of the required 30-50kHz.

Above calculated values:

$$C_{out,eff} = 44\mu F, \quad L = 1.84\mu H, \quad R_o = \frac{V_{out}}{I_{out}} = 0.36\Omega$$

The LC double-pole frequency was:

$$f_{LC} = \frac{1}{2\pi\sqrt{LC_{out}}}$$

$$f_{LC} = 17.67kHz$$

The output capacitor ESR zero was:

$$f_{ESR} = \frac{1}{2\pi C_{out} ESR}$$

$$f_{ESR} = 1.21MHz$$

The compensation values are then:

$$R_{C1} = 7.24k\Omega$$

$$C_{C1} = 2.49nF$$

$$C_{C2} = 91.1pF$$

$$R_{C2} = 297.5\Omega$$

$$C_{C3} = 443.7pF$$

Which fit the design spec.

3. SPICE Simulations

After completing the initial hand calculations, simulations were used to check the expected behavior of the converter before building and testing the PCB.

LTspice was used to simulate the converter transient behavior. However, LTspice was not used for the final frequency response verification because the average model in LTspice gave a phase margin close to 0 degrees (usually outputting somewhere around +- 3deg at crossover.) However, the transient response was stable with relatively minimal ringing, which indicates that the average model would be wrong (that little phase margin would be only positive feedback resulting in only ringing.) Because of this, TI WEBENCH Power Simulator was used to model the average response and get phase/gain margin values.

The inductor ripple current was first checked at full load. The simulated inductor current varied from approximately 4.42A to 5.58A, giving a peak-to-peak ripple current of about 1.16A. This is below the maximum allowed ripple current of 30% of the 5A full-load current, which is 1.5A.

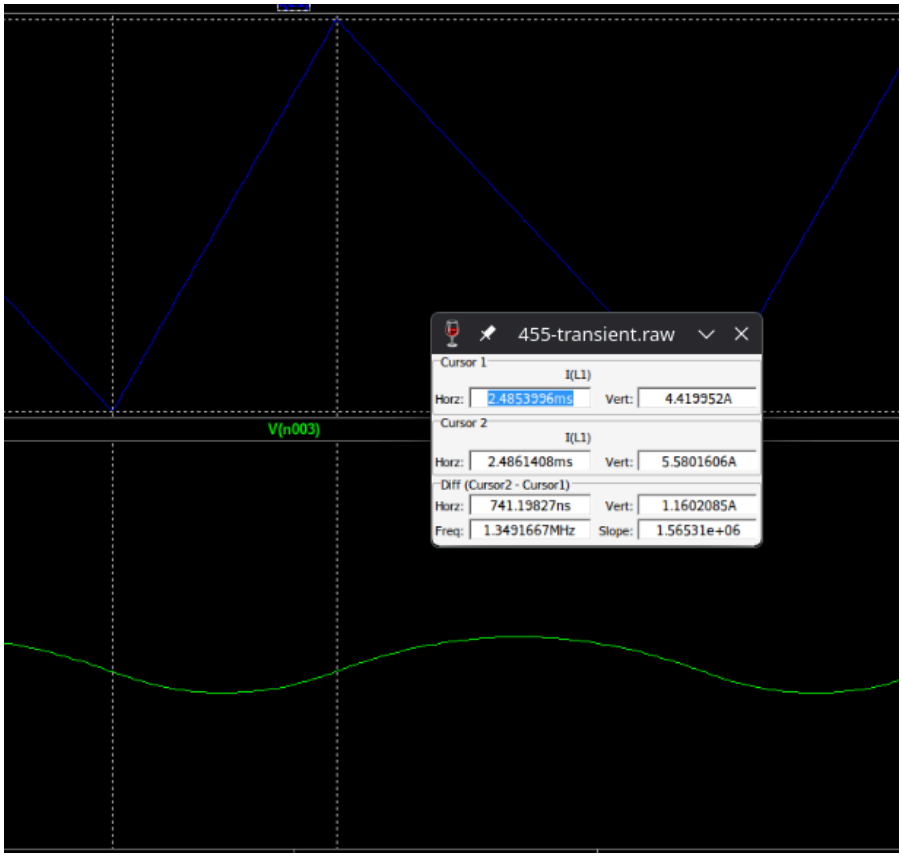


Figure 3: Inductor current Ripple

The output voltage ripple was also checked during steady-state operation. The simulated output voltage ripple was approximately 2.68mV peak-to-peak. This is well below the 1% output ripple requirement. Since 1% of 1.8V is 18mV, the simulated ripple result satisfies the design requirement.

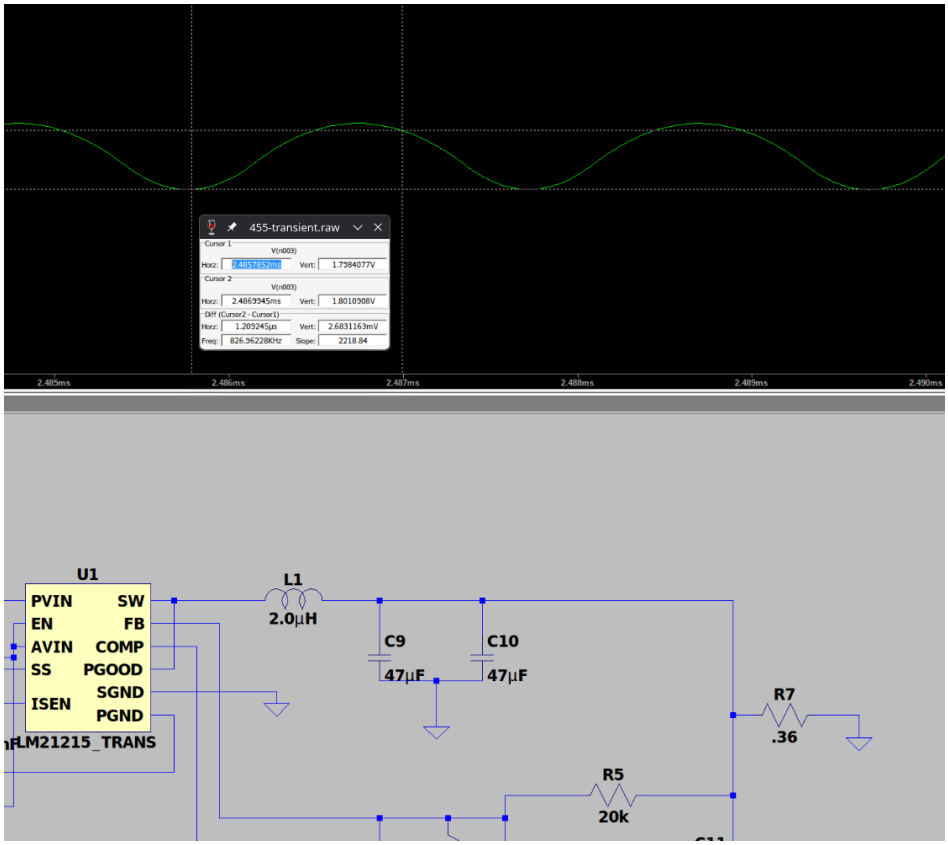


Figure 4: Vout Ripple

Next, transient responses were tested.

For a 2A to 3A load-step case, the output voltage droop was approximately 48.7mV. This is slightly under the project limit of 50mV, so the design meets the transient droop requirement in simulation.

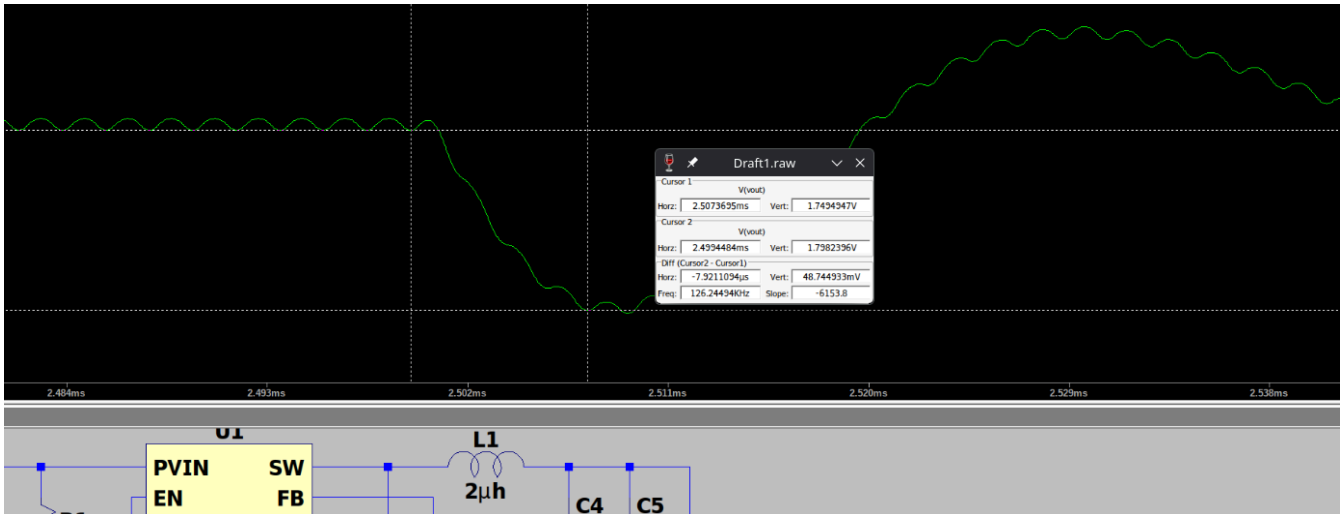


Figure 5: Transient within Vdroop spec

Similarly, the transient response showed quick damping, within the "3-ring" heuristic mentioned in class:

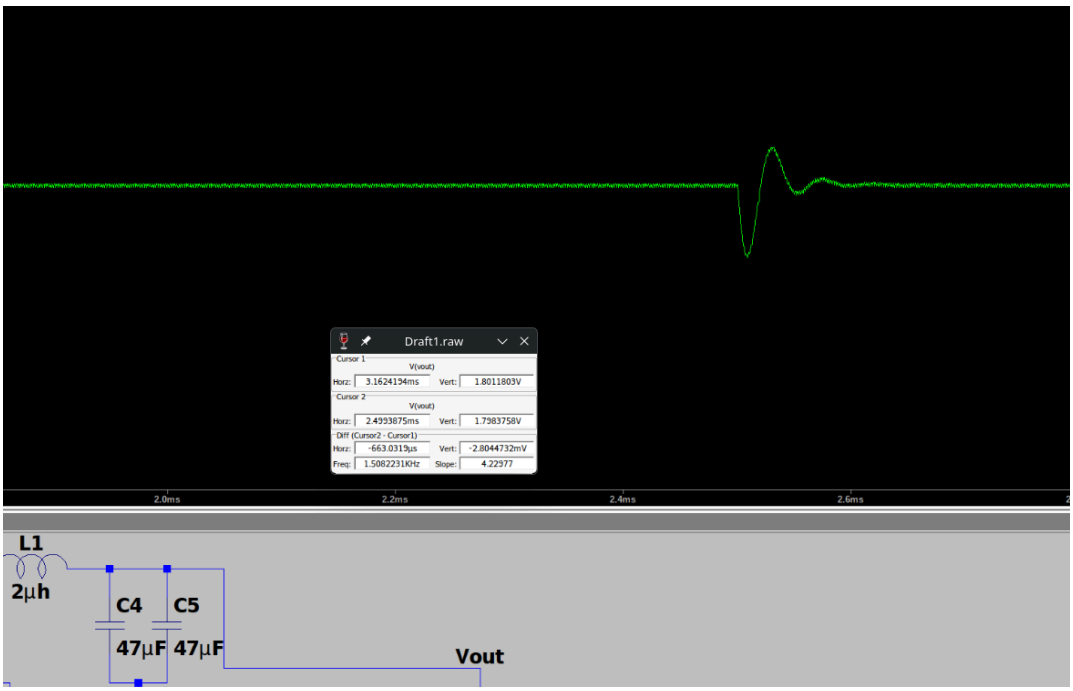


Figure 5: Transient rings

Here, the values needed to be adjusted to meet the transient droop specification. Originally, with the values calculated above, the ringing was quickly damped, suggesting that the converter was stable. However, the voltage droop was still well above the specification at approximately 180 mV. This suggests that the compensation network had enough damping for stability, but the transient response was too slow and the output capacitance was not providing enough charge during the load step.

To improve this, the output capacitance was increased and the compensation network values were adjusted. The inductor was also changed to a practical 2.0uH value (especially because most inductors on the market had a +-10-20% tolerance, making 2 within range of the calculated 1.84 anyways.) Increasing the output capacitance from 44 to 94 helps reduce the immediate voltage drop during the load transient, while adjusting the compensation network improves how quickly the control loop responds and recovers back to 1.8V. After these changes, the simulated 2A to 3A load step showed a voltage droop of approximately 48.7mV, which is just under the 50mV requirement.

Finally, to ensure that the design was within the phase margin and crossover frequency spec, TI WEBENCH was used to simulate the design with the new calculated values. This simulation gave a crossover frequency of approximately 26kHz and a phase margin of about 61 degrees. Since the target crossover frequency was in the 30kHz to 50kHz range and the phase margin requirement was greater than 50degrees, this result indicated that the compensation network should be stable.

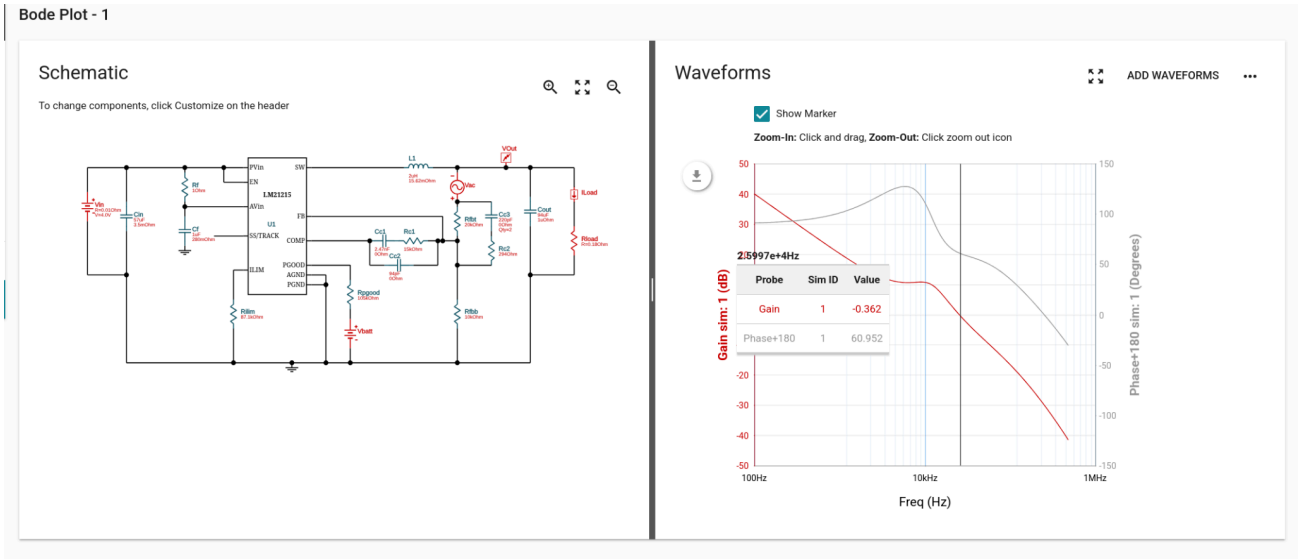


Figure 6: Compensation Phase & Gain margin

The crossover is below the 30kHz minimum value because the output capacitance was increased to meet droop requirements. Since the simulation stayed stable, I figured it was one or the other and didn't want to waste anymore time trying to get within both, and just ordered the parts.

4. Layout

Layout, while mostly just following the designed schematic, needed to ensure 2 things: large copper pours were used for any of the power stage components, compensation loop components were placed somewhat at a distance from the inductor to minimize EMF interference, and

input/output caps were placed as close to the IC as possible to ensure that parasitic capacitance was minimized. The final layout can be seen below:

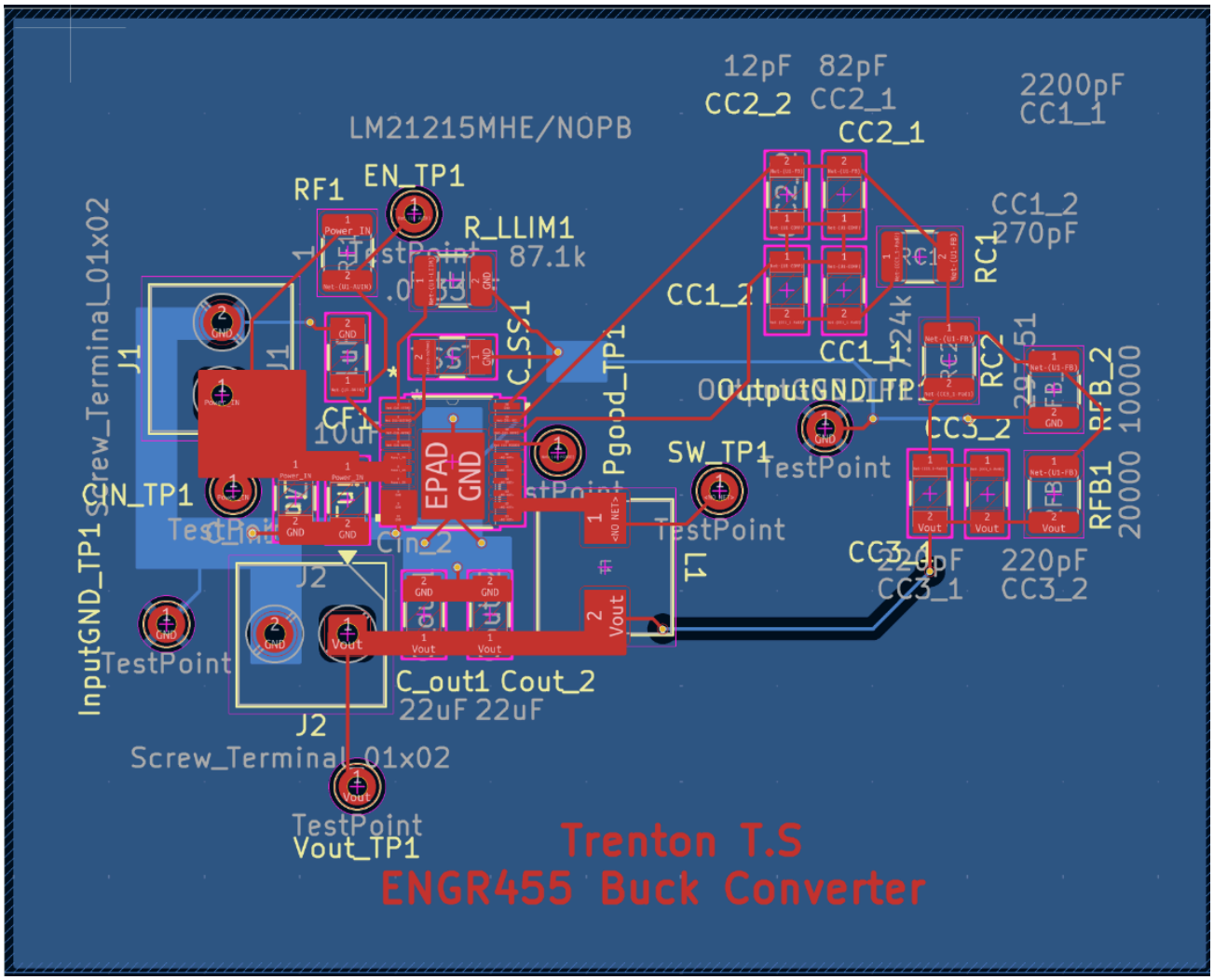


Figure 7: PCB Layout

5. Testing

Finally, testing. While I would like to say that everything went according to plan and my tests roughly matched my design simulations, this did not account for human error (idiocy would be a better description.) I lost my 82pF CC2 compensation capacitors.

In the intended design, CC2 was made from two capacitors in parallel:

CC2 = 82pF + 12pF = 94pF, and since the 82pF capacitors were lost, the actual assembled value became: CC2 = 12pF + 12pF = 24pF.

This means that the high-freq compensation pole was significantly moved into higher frequencies, from roughly 176kHz, to 1.33MHz

Because this pole moved above the 500kHz switching frequency, the compensation network no longer attenuated high-frequency switching noise as strongly as intended. This did not completely destroy regulation, since the crossover frequency and main LC compensation zeros were still mostly set by RC1, CC1, RC2, and CC3. However, it did make the real hardware more sensitive to switching noise, ringing, and layout parasitics than the simulated design.

The converter still produced approximately the correct DC output voltage which showed it was still within some stable margin, but the results were expected to show more ringing and high-frequency noise, which is what was seen.

I first measured DC Voltage - the measured output voltage was approximately 1.7838V. This is close to the 1.8V design target and confirms that the feedback divider and regulator were operating correctly.

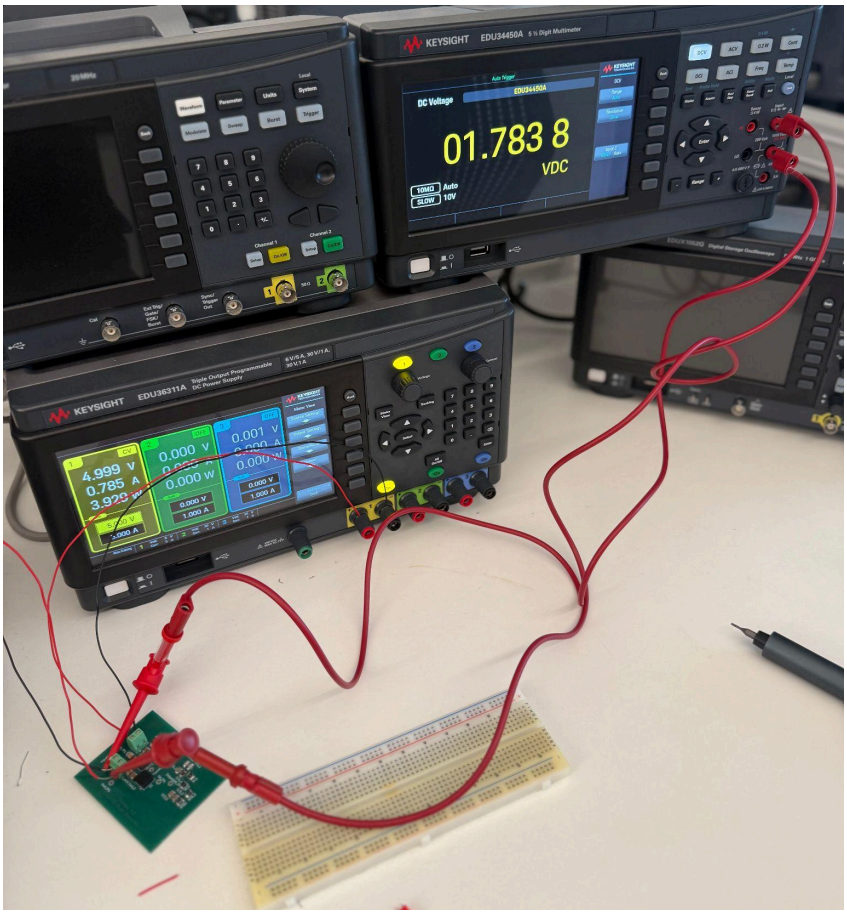


Figure 8: DC output

The output ripple was then measured at full load. The oscilloscope measured approximately 40mV peak-to-peak output ripple. This is higher than the original simulated ripple, but it is reasonable given the real PCB parasitics, measurement noise, and the missing compensation capacitor.

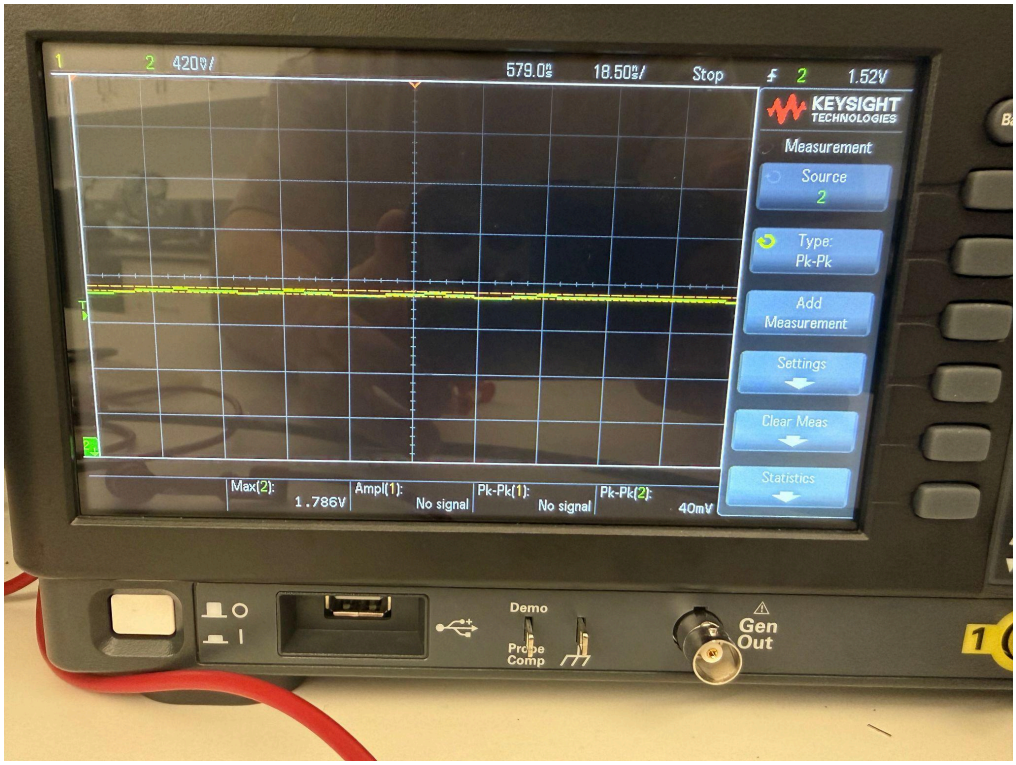


Figure 9: Output Ripple

Next, the switch node waveform was also measured to check the value of the duty cycle. The measured interval was x1-x2=0.758us. And given the actual switching freq of 500kHz = 2us period, this gives a duty of 0.758/2us, resulting in a duty of ~35.7%. This matches the calculated 36% expected.

*note: I forgot i could just use the “measure” button on the oscilloscope to get the duty which is why this is calculated by hand.

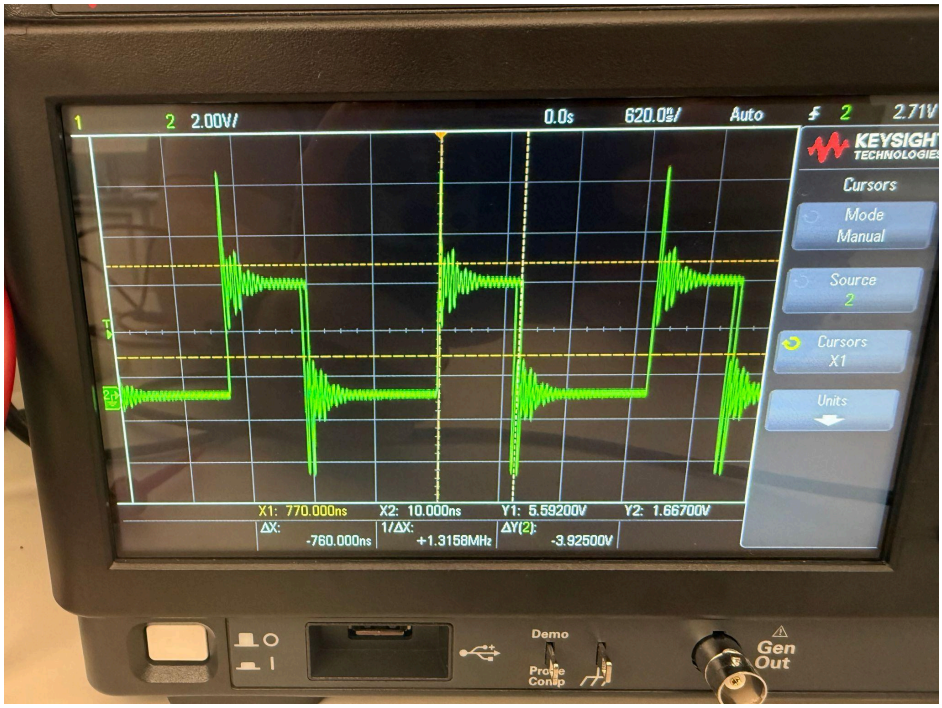


Figure 10: Duty Measurement

The inductor current waveform was measured using a current probe. The measured inductor ripple current was approximately 950mA peak-to-peak. This is below the maximum ripple requirement of 30% of the 5A full-load current:

30% of 5A = 1.5A; Since 950 mA is less than 1.5A, the inductor ripple current met the requirement.

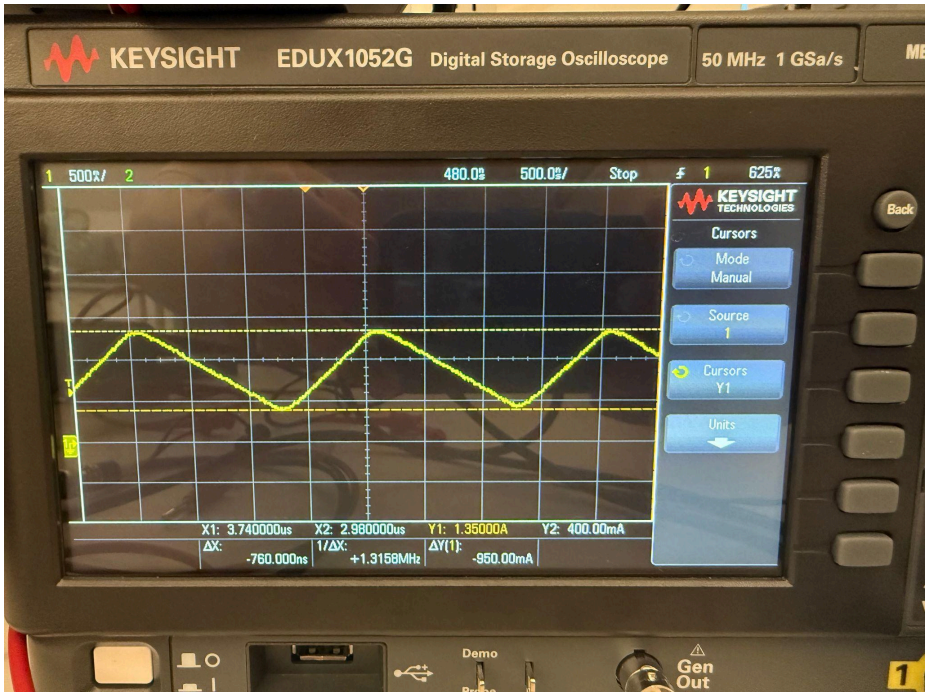


Figure 11: Inductor Current & Ripple

Transient testing was then performed using 2A to 3A and 3A to 2A load steps. For the 2A to 3A load step, the oscilloscope measured approximately 510mV peak-to-peak ringing. For the 3A to 2A load step, the oscilloscope measured approximately 350mV peak-to-peak ringing. These results show that the output responded to the load step, but the real hardware had significantly more ringing than the simulation, as well as significantly more drooping.



Figure 12, 13: 3A-2A & 2A-3A Transient

The extra ringing is most likely due to a combination of the changed compensation network, PCB parasitics, and measurement setup. The missing 82pF CC2 capacitor moved the high-frequency compensation pole upward, reducing high-frequency attenuation. As a result, the loop was less

effective at suppressing fast switching noise and transient ringing. Also, as with basically all real-world testing, things like ESR of the components, parasitics of traces, bad soldering, among other things can easily contribute to noise as shown here.

Overall, the converter successfully regulated near the desired 1.8V output and met the inductor ripple requirement. However, the measured output ripple and transient ringing were wayyy worse than expected from simulation. The most likely explanation is that the actual assembled compensation network did not match the simulated design because CC2 was reduced from 94pF to 24pF, causing the real converter to have less high-frequency damping than intended.

6. Conclusion

In this report, I describe the path in designing and creating a 5V-1.8V DC-DC buck converter using the TI LM21215 buck IC. I first go through hand calculations for major components in the design. I then describe the initial simulations done to ensure that the calculated values were right, and describe what changes were made for the final design values. I then show the PCB layout and what things were considered when doing so. Finally, I show the output testing and describe what likely lead to the wildly different output values that were recorded.

While the final design did do what was intended - regulating 5V down to 1.8V, there were obviously a lot of things that could've gone better. The main one being not procrastinating on the testing, and realizing that I had lost my component, and as such, having time to order a new one from DigiKey. Otherwise, I learned a lot in this project, in everything from design to actually soldering things, and going forward, would be confident in designing another buck regulator.